

EDUCATION	PhD in Physics at CUNY Graduate Center, New York, NY	2016–2022
	Dissertation: <i>Collapse of Two-Dimensional Magnetic Skyrmions</i>	
	B.S. in Physics and Mechanical Engineering at Tufts University, Medford, MA	2011–2015
RESEARCH EXPERIENCE	Postdoctoral Fellow , Columbia Climate School Postdoctoral Fellowship	Sept. 2024–Present
	– Lead research projects on atmospheric convection’s relationship with the global energy cycle – Develop novel methods for interpreting clouds with remote sensing data, addressing retrieval gaps	
	Postdoctoral Researcher , The City College of New York, CUNY	Sept. 2022–Aug. 2024
	– Combined geostationary satellite data with a basic cloud model to quantify vertical mass flux in the tropics – Redesigned programming algorithm to achieve major speed reduction, enabling large-scale deployment	
	Graduate Student Intern , NASA Goddard Institute for Space Studies	Oct. 2021–Aug. 2022
	Part-time role (10 hours/week) – Collaborated with a senior scientist, high school teacher, undergraduate student, and high school student – Analyzed several decades of satellite data to assess differences in land surface temperature in New York City	
MANUSCRIPTS IN PREP	Graduate Student Researcher , CUNY Graduate Center	July 2017–Sept. 2022
	– Developed classical and quantum theories to model two-dimensional spin systems, informing experiments – Parallelized thousands of computations using both multiprocessing and multithreading, validating theories	
	Derras-Chouk, Amel, Gregory Elsaesser, and Zhengzhao Johnny Luo (In prep). <i>The Diurnal Cycle of Deep Convective Precipitation Efficiency in the Tropics</i> .	
PUBLICATIONS	Derras-Chouk, Amel, Gregory Elsaesser, Zhengzhao Johnny Luo, Toshi Matsui, Andreas F. Prein, and Jingbo Wu (2026). “Estimating Vertical Velocity in Convective Updrafts from Temperature, Pressure, and Latent Heating”. In: <i>Journal of the Atmospheric Sciences</i> , e250141.	
	Braneon, Christian, Luis Ortiz, Daniel Bader, Naresh Devineni, Philip Orton, Bernice Rosenzweig, Timon McPhearson, Lauren Smalls-Mantey, Vivien Gornitz, Talea Mayo, Sanketa Kadam, Hadia Sheerazi, Equisha Glenn, Liv Yoon, Amel Derras-Chouk, Joel Towers, Robin Leichenko, Deborah Balk, Peter Marcotullio, and Radley Horton (2024). “NPCC4: New York City climate risk information 2022—observations and projections”. In: <i>Annals of the New York Academy of Sciences</i> 1539.1, pp. 13–48.	
	Derras-Chouk, Amel and Zhengzhao Johnny Luo (2024). “A geostationary satellite-based approach to estimate convective mass flux and revisit the hot tower hypothesis”. In: <i>Surveys in Geophysics</i> 45.6, pp. 1959–1977.	
	Derras-Chouk, Amel, Eugene M Chudnovsky, and Dmitry A Garanin (2022). “Dynamics of the collapse of a ferromagnetic skyrmion in a centrosymmetric lattice”. In: <i>Physical Review B</i> 105.13, p. 134432.	
	Derras-Chouk, Amel and Eugene M Chudnovsky (2021). “Skyrmions near defects”. In: <i>Journal of Physics: Condensed Matter</i> 33.19, p. 195802.	
	Derras-Chouk, Amel, Eugene M Chudnovsky, and Dmitry A Garanin (2021). “Quantum states of a skyrmion in a two-dimensional antiferromagnet”. In: <i>Physical Review B</i> 103.22, p. 224423.	

- (2019). “Thermal collapse of a skyrmion”. In: *Journal of Applied Physics* 126.8.
- (2018). “Quantum collapse of a magnetic skyrmion”. In: *Physical Review B* 98.2, p. 024423.

Derras-Chouk, Amel, Eugene M Chudnovsky, Dmitry A Garanin, and Reem Jaafar (2018). “Graphene cantilever under Casimir force”. In: *Journal of Physics D: Applied Physics* 51.19, p. 195301.

TEACHING EXPERIENCE	Ajunct Lecturer Lehman College Aug. 2017–May 2021	
	– Lab Section of General Physics 1 (PHY 166/168)	
	– Lab Section of General Physics 2 (PHY 167/169)	
SELECTED PRESENTATIONS	Jan. 2026	American Meteorological Society Annual Meeting 2026, Houston, TX, USA. <i>Estimating vertical velocity from temperature, pressure, and latent heating</i>
	Dec. 2025	American Geophysical Union Annual Meeting, New Orleans, LA, USA. <i>The diurnal cycle of deep convective precipitation efficiency in the tropics</i>
	Nov. 2025	Invited talk at INCUS Science Team Meeting, Fort Collins, CO, USA. <i>Towards Developing a Historical Record of Updraft Vertical Velocity</i>
	Oct. 2025	Columbia Climate School Research Symposium, New York, NY, USA. <i>How fast are thunderstorm updrafts?</i>
	Sept. 2025	Invited seminar at NOAA Geophysical Fluid Dynamics Laboratory, Princeton, NJ, USA. <i>Quantifying Vertical Motion Using Satellite Data.</i>
	July 2025	Gordon Research Conference on Atmospheric Radiation (Poster presentation), Lewiston, ME, USA. <i>Estimating vertical velocity from temperature, pressure, and latent heating.</i>
	Apr. 2025	NASA Goddard Institute for Space Science, New York, NY, USA. <i>Quantifying Convective Vertical Velocity and Precipitation Efficiency Using Satellite Data.</i>
	Feb. 2025	CloudSat Calipso Science Team Meeting, Fort Collins, CO, USA. <i>Estimating convective mass flux, detrainment, and precipitation efficiency using CloudSat, Geostationary, and IMERG data.</i>
	Dec. 2023	AGU Annual meeting (Poster presentation), San Francisco, CA. <i>A Satellite-Based Approach to Estimating Convective Mass Flux and Revisiting the Hot Tower Hypothesis</i>
COMMUNITY SERVICE	Volunteer Word Up Community Bookshop/Libreria Comunitaria Sept. 2017–Dec. 2023	
	– Led weekly shifts in a volunteer-run bookstore, managing daily sales and accounting	
	– Developed a donation tracking program to automate data management, designing for non-technical users	
HONORS AND AWARDS	Columbia Climate School Postdoctoral Fellowship	2024–Present
	CUNY Graduate Center Dissertation Fellowship	2020–2021
	Physics Tithe Fellowship	2018–2022